Two of the five equal parts in the figure are coloured. The coloured part is $\frac{2}{5}$ of the whole figure.

When two lines are drawn horizontally in the same figure, the figure gets divided into 15 equal parts. So, the fraction that shows the coloured part now has changed. Now, the fraction of the coloured part is $\frac{6}{15}$ but coloured size is the same.

Two figures that have the same size are called equivalent fractions.

Therefore, we see that $\frac{2}{5} = \frac{6}{15}$

When the numerator and denominator of a fraction are multiplied by the same non-zero number, we get a fraction that is equivalent to the given fraction.

Example 6.4

Find a fraction with denominator 20 which is equivalent to $\frac{2}{5}$.

Solution

$\frac{2}{5} = \frac{\square}{20}$. We must find the right number for the box.

Here, 4 times the denominator 5 is 20. So, multiply the numerator also by 4

Therefore,

$$\frac{2}{5} = \frac{2 \times 4}{5 \times 4} = \frac{8}{20}$$

Hence, the fraction $\frac{8}{20}$ is equivalent to $\frac{2}{5}$.

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